

Phase Weight Rationale

Weight order: (corr, fmt, parse, finish, tol, brev). This matches the reward expression order: correctness, formatting, parseability, finished, tolerance, brevity. These are phase initial weights; the controller may update them after checkpoint evaluation.

Phase	Configured Goal	Stage Mix	Initial Weights	Rationale
phase_a	Structure stabilization on Stage 1.	stage1_easy_numeric: 100%	(2, 1, 1, 1.5, 0, 0.25)	Structure-first start: formatting, parseability, and completion weights are high enough to make tagged, parseable, completed answers the early objective before stronger correctness pressure.
phase_b	Correctness strengthening with Stage 1/2 mix.	stage1_easy_numeric: 70%, stage2_float_numeric: 30%	(4, 0.75, 0.75, 1, 0, 0.2)	Moves pressure toward correctness after Phase A; formatting and parseability remain active as guard rails while Stage 2 enters the mix.
phase_c	Precision and harder reasoning with Stage 2/3 mix.	stage2_float_numeric: 60%, stage3_hard_numeric: 40%	(5, 0.5, 0.5, 0.75, 1, 0.2)	Raises correctness for harder numeric reasoning and enables tolerance reward for near-miss numeric answers; structure weights are lower because controller guards maintain formatting.

Component bounds: correctness 1.0–8.0; format and parseable 0.25–2.0; finished 0.5–2.0; tolerance 0.0–2.0; brevity 0.0–1.0.